

Weak type $(1, 1)$ for variation of the Laguerre heat semigroup

A full answer to the endpoint question in arXiv:2110.03493

Autonomous mathematical discovery run `fa_banach_001`, lane 5

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Status: candidate full, likely valid. The proof is analytic. A short exact symbolic script only audits the degrees of three explicitly displayed polynomials.

1 Source question and answer

Betancor and de León-Contreras study variation operators for Laguerre polynomial expansions in [1]. On page 9, while proving their Poisson endpoint theorem, they state that they do not know the weak type $(1, 1)$ bound for the variation of the transformed Laguerre heat semigroup $\{\widehat{W}_t^\alpha\}_{t>0}$; see Figure 1.

VARIATION INEQUALITIES

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2.1.2. $\mathcal{V}_\rho(\{t^\beta \mathcal{D}_t^\beta \hat{P}_t^\alpha\}_{t>0})$ is bounded from $L^1((0, \infty)^n, \nu_\alpha)$ into $L^{1,\infty}((0, \infty)^n, \nu_\alpha)$.

In this moment we do not know how to see that the operator $\mathcal{V}_\rho(\{\hat{W}_t^\alpha\}_{t>0})$ is bounded from $L^1((0, \infty)^n, \nu_\alpha)$ into $L^{1,\infty}((0, \infty)^n, \nu_\alpha)$, so we will obtain our result by using a procedure developed by Sasso in [46] and [49].
Observe that

$$\begin{aligned} \hat{W}_t^\alpha(x, y) &= (1 - e^{-t})^{-n} \exp\left(-\frac{e^{-t}}{1 - e^{-t}} \sum_{i=1}^n (x_i^2 + y_i^2)\right) \prod_{i=1}^n \Gamma(\alpha_i + 1) (e^{-t/2} x_i y_i)^{-\alpha_i} \\ &\quad \cdot I_{\alpha_i}\left(\frac{2e^{-t/2} x_i y_i}{1 - e^{-t}}\right), \quad x = (x_1, \dots, x_n), y = (y_1, \dots, y_n) \in (0, \infty)^n \text{ and } t > 0. \end{aligned}$$

According to [32, (5.10.22)],

$$I_\nu(z) = \frac{\left(\frac{z}{2}\right)^\nu}{\sqrt{\pi} \Gamma(\nu + 1/2)} \int_{-1}^1 (1 - s^2)^{\nu-1/2} e^{-sz} ds, \quad z \in (0, \infty), \quad \nu > -1/2,$$

we can write, for every $x, y \in (0, \infty)^n$ and $t > 0$,

$$\hat{W}_t^\alpha(x, y) = (1 - e^{-t})^{-n} \exp\left(-\frac{e^{-t}}{1 - e^{-t}} \sum_{i=1}^n (x_i^2 + y_i^2)\right) \prod_{i=1}^n \Gamma(\alpha_i + 1) (e^{-t/2} x_i y_i)^{-\alpha_i} \cdot I_{\alpha_i}\left(\frac{2e^{-t/2} x_i y_i}{1 - e^{-t}}\right).$$

Figure 1: The endpoint question and the heat-kernel formula in the source paper [1, p. 9].

Let $n \geq 1$, $\alpha = (\alpha_1, \dots, \alpha_n) \in (0, \infty)^n$, and

$$d\nu_\alpha(x) = 2^n \prod_{j=1}^n \frac{x_j^{2\alpha_j+1} e^{-x_j^2}}{\Gamma(\alpha_j + 1)} dx_j.$$

For $\rho > 2$, write

$$\mathcal{V}_\rho(\{T_t\}_{t>0})f(x) = \sup_{0 < t_k < \dots < t_1} \left(\sum_{j=1}^{k-1} |T_{t_j} f(x) - T_{t_{j+1}} f(x)|^\rho \right)^{1/\rho}.$$

Theorem 1. For every $\rho > 2$ there is $C = C(n, \alpha, \rho)$ such that, for every $f \in L^1((0, \infty)^n, \nu_\alpha)$ and $\lambda > 0$,

$$\nu_\alpha\left(\left\{x : \mathcal{V}_\rho(\{\widehat{W}_t^\alpha\}_{t>0})f(x) > \lambda\right\}\right) \leq \frac{C}{\lambda}\|f\|_{L^1(\nu_\alpha)}.$$

Consequently the same estimate holds for the original Laguerre polynomial heat semigroup on $L^1(\mu_\alpha)$, by the isometry $x \mapsto x^2$ used in [1, Section 2.1].

The point is that nearly all endpoint machinery needed for the heat family is already present inside the source's proof for the subordinated Poisson family. The missing local calculation is supplied below.

2 Scalar components and bounded total variation

Put

$$A = n + \sum_{j=1}^n \alpha_j, \quad X = |x|^2, \quad Y = |y|^2, \quad Z = \sum_{j=1}^n x_j y_j s_j$$

and, for $r \in (0, 1)$,

$$G_{x,y,s}(r) = (1 - r^2)^{-A} \exp\left[-\frac{Xr^2 + Y - 2Zr}{1 - r^2}\right]. \quad (1)$$

With $r = e^{-t/2}$, the source's integral representation is

$$\widehat{W}_t^\alpha(x, y) = e^{|y|^2} \int_{(-1,1)^n} G_{x,y,s}(e^{-t/2}) \Pi_\alpha(s) ds, \quad (2)$$

where

$$\Pi_\alpha(s) = \prod_{j=1}^n \frac{\Gamma(\alpha_j + 1)}{\Gamma(\alpha_j + 1/2)\sqrt{\pi}} (1 - s_j^2)^{\alpha_j - 1/2}.$$

We use the elementary fact that if a differentiable real function has at most d critical points, then its total variation is at most $2(d + 1)$ times its supremum norm. The same conclusion holds when the derivative numerator below is identically zero.

Lemma 1 (Uniform critical-point bounds). For fixed x, y, s , off the kernel diagonal,

$$\text{Var}_{0 < r < 1} G_{x,y,s}(r) \leq 8 \sup_{0 < r < 1} G_{x,y,s}(r), \quad (3)$$

$$\text{Var}_{0 < r < 1} \partial_{x_i} G_{x,y,s}(r) \leq 12 \sup_{0 < r < 1} |\partial_{x_i} G_{x,y,s}(r)|, \quad (4)$$

$$\text{Var}_{0 < r < 1} \partial_{y_i} G_{x,y,s}(r) \leq 10 \sup_{0 < r < 1} |\partial_{y_i} G_{x,y,s}(r)|. \quad (5)$$

These are also the total variations under the monotone parametrization $r = e^{-t/2}$, $t \in (0, \infty)$.

Proof. Logarithmic differentiation of (1) gives

$$\frac{dG}{dr} = \frac{2G}{(1 - r^2)^2} P_0(r), \quad P_0(r) = Z + (A - X - Y)r + Zr^2 - Ar^3. \quad (6)$$

Thus G' has at most three zeros unless $P_0 \equiv 0$, proving (3). Moreover

$$\partial_{x_i} G = -\frac{2(r^2 x_i - r y_i s_i)}{1 - r^2} G, \quad \partial_{y_i} G = -\frac{2(y_i - r x_i s_i)}{1 - r^2} G. \quad (7)$$

After differentiating these expressions in r and multiplying by the positive factor $(1 - r^2)^3/G$, the two derivative numerators are polynomials of degrees at most five and four, respectively. Hence they have at most five and four zeros unless identically zero. The total-variation observation proves (4)–(5). The exact expanded polynomials and degree checks are reproduced by `code/verify_derivative_degrees.py`. $\square$

Since $\rho \geq 1$, every differentiable scalar function h satisfies

$$\|h\|_{E_\rho} \leq \int_0^\infty |h'(t)| dt. \quad (8)$$

Lemma 1 turns this crude bound into sharp enough kernel estimates.

3 Global part

We use exactly the source's cutoff. Let

$$q_\pm(x, y, s) = |x|^2 + |y|^2 \pm 2 \sum_{j=1}^n x_j y_j s_j$$

and choose a smooth $0 \leq \varphi \leq 1$ which equals 1 on

$$N_1 = \left\{ q_-(x, y, s)^{1/2} \leq \frac{C_0}{1 + |x| + |y|} \right\}$$

and vanishes outside the corresponding enlarged region N_2 . Decompose $\widehat{W}_t^\alpha = \widehat{W}_{t,\text{loc}}^\alpha + \widehat{W}_{t,\text{glob}}^\alpha$ by inserting φ and $1 - \varphi$ into (2).

The proof of Lemma 2.1 in [1] establishes, before any Poisson-specific step is used, the following estimate for the heat kernel components:

$$\int_0^\infty |\partial_t \widehat{W}_{t,\text{glob}}^\alpha f(x)| dt \leq \mathbb{K}_\alpha(|f|)(x). \quad (9)$$

Indeed, (3) bounds the time integral by a fixed multiple of $\sup_t G_{x,y,s}(e^{-t/2})$. On the support of $1 - \varphi$, the source proves

$$\sup_t G_{x,y,s}(e^{-t/2}) \lesssim K_\alpha(x, y, s) := \begin{cases} e^{-|y|^2}, & Z \leq 0, \\ \left(\frac{q_+}{q_-}\right)^{A/2} \exp\left[-\frac{|y|^2 - |x|^2}{2} - \frac{\sqrt{q_+ q_-}}{2}\right], & Z > 0. \end{cases} \quad (10)$$

The positive integral operator formed from $K_\alpha(1 - \varphi)$ against $dm_\alpha(y) = e^{|y|^2} d\nu_\alpha(y)$ is $\mathbb{K}_\alpha$. By the Sasso estimates invoked and reproduced in [1, Lemma 2.1], it is weak type $(1, 1)$ and strong type (p, p) for $1 < p < \infty$ with respect to ν_α .

Combining (8) and (9),

$$\mathcal{V}_\rho(\{\widehat{W}_{t,\text{glob}}^\alpha\}_{t>0})f \leq \mathbb{K}_\alpha(|f|), \quad (11)$$

so the global variation is weak $(1, 1)$ and strong L^2 .

4 Local vector-valued Calderón–Zygmund estimate

Let $H_t^\alpha(x, y)$ be the local kernel, now viewed against the doubling measure m_α :

$$H_t^\alpha(x, y) = \int_{(-1,1)^n} G_{x,y,s}(e^{-t/2}) \varphi(x, y, s) \Pi_\alpha(s) ds.$$

The standard local comparison estimate used in [1, Lemma 2.2] implies, on N_2 ,

$$\sup_t G_{x,y,s}(e^{-t/2}) \lesssim q_-(x, y, s)^{-A}, \quad (12)$$

$$\sup_t (|\nabla_x G_{x,y,s}(e^{-t/2})| + |\nabla_y G_{x,y,s}(e^{-t/2})|) \lesssim q_-(x, y, s)^{-A-1/2}. \quad (13)$$

For completeness, (12) follows by comparing $q_-(rx, y, s)$ with $q_-(x, y, s)$ and maximizing $u^{-A}e^{-cq_-/u}$. For (13), use $|rx_i - y_i s_i| \leq q_-(rx, y, s)^{1/2}$ (and the analogous y estimate), absorb the square root into half of the exponential, and maximize once more.

Equations (8), (3)–(5), (12)–(13), and $|\nabla\varphi| \lesssim q_-^{-1/2}$ give

$$\|H^\alpha(x, y)\|_{E_\rho} \lesssim \int_{(-1,1)^n} q_-(x, y, s)^{-A} \Pi_\alpha(s) ds \lesssim \frac{1}{m_\alpha(B(x, |x-y|))}, \quad (14)$$

$$\begin{aligned} \|\nabla_x H^\alpha(x, y)\|_{E_\rho} + \|\nabla_y H^\alpha(x, y)\|_{E_\rho} &\lesssim \int_{(-1,1)^n} q_-(x, y, s)^{-A-1/2} \Pi_\alpha(s) ds \\ &\lesssim \frac{1}{|x-y| m_\alpha(B(x, |x-y|))}. \end{aligned} \quad (15)$$

The final integral estimates are precisely the Bessel-kernel estimates used in [1, Lemma 2.2]. Thus the local heat kernel satisfies the same E_ρ -valued Calderón–Zygmund bounds as the source’s Poisson kernel.

It remains to supply an L^2 input. The family $\{\widehat{W}_t^\alpha\}$ is a Stein symmetric diffusion semigroup, so the strong variation theorem of Le Merdy and Xu [2, Corollary 6.1] yields

$$\|\mathcal{V}_\rho(\{\widehat{W}_t^\alpha\})f\|_{L^2(\nu_\alpha)} \lesssim \|f\|_{L^2(\nu_\alpha)}.$$

The global part is strong L^2 by (11); hence the local part is strong L^2 as well.

Now repeat the source’s measure-transfer and vector-valued Calderón–Zygmund argument (its Lemma 2.3 and the paragraph following it). The local support of φ transfers the L^2 estimate between ν_α and the doubling measure m_α . Bounds (14)–(15) then give weak $(1, 1)$ on m_α , uniformly for variation restricted to $t \in [1/N, N]$. The same local transfer gives weak $(1, 1)$ on ν_α . Letting $N \rightarrow \infty$ by monotone convergence proves

$$\mathcal{V}_\rho(\{\widehat{W}_{t,\text{loc}}^\alpha\}_{t>0}) : L^1(\nu_\alpha) \longrightarrow L^{1,\infty}(\nu_\alpha). \quad (16)$$

Combining (11) and (16) proves Theorem 1.

5 Audit, novelty boundary, and limitations

Run the exact algebra audit from the repository root:

```
conda run --no-capture-output -n sandbox python \
  runs/fa_banach_001/solutions/full/\
  2110.03493_laguerre_heat_variation_weak_11/code/\
  verify_derivative_degrees.py
```

Its captured output is included beside the script. The computation is not a numerical experiment: it expands formal derivatives and asserts exact polynomial identities.

A bounded novelty search on 12 August 2026 covered the run indexes; the exact arXiv id and question sentence; combinations of “Laguerre heat semigroup,” “variation,” and “weak type $(1, 1)$ ”; and later endpoint papers citing the source. The 2012 Laguerre variation paper [5] gives strong- L^p and Hardy-space results in related Laguerre settings. The 2024 endpoint paper [6] gives $H^1 \rightarrow L^1$ and $L^\infty \rightarrow \text{BMO}$ estimates. Neither states this $L^1 \rightarrow L^{1,\infty}$ heat-variation theorem. Later Ornstein–Uhlenbeck work proves analogous Gaussian results but does not cover arbitrary Laguerre parameters. No exact prior answer was found; keyword and citation searches remain incomplete, so novelty requires expert review.

The proof claims only $\rho > 2$ and $\alpha \in (0, \infty)^n$, exactly the range in which the source develops the needed global Sasso estimate. It makes no claim at $\rho = 2$, for negative Laguerre parameters, or for dimension-free constants. The main review-sensitive point is that the local spatial-gradient total variations, not merely the scalar-kernel variation, are required; Lemma 1 supplies this missing step explicitly.

Human-review recommendation. Check the factorization in (6) and the two degree bounds following (7); then compare (9)–(10) and (12)–(15) line by line with Lemmas 2.1–2.3 of the source. Those checks isolate every nonstandard ingredient.

References

- [1] J. J. Betancor and M. de León-Contreras, *Variation inequalities for Riesz transforms and Poisson semigroups associated with Laguerre polynomial expansions*, arXiv:2110.03493v2 (2022); Anal. Appl. 22 (2024), 1–56. <https://arxiv.org/abs/2110.03493>.
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- [3] E. Sasso, *Spectral multipliers of Laplace transform type for the Laguerre operator*, Bull. Austral. Math. Soc. 69 (2004), 255–266.
- [4] E. Sasso, *Weak type estimates for Riesz–Laguerre transforms*, Bull. Austral. Math. Soc. 75 (2007), 397–408.
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- [6] J. J. Betancor, E. Dalmasso, P. Quijano, and R. Scotto, *Endpoint estimates for harmonic analysis operators associated with Laguerre polynomial expansions*, Math. Nachr. 297 (2024), 2365–2389; arXiv:2210.14394.